

Impact of different stages of juice processing on the anthocyanin, flavonol, and procyanidin contents of cranberries.



Juice is the most common form in which cranberries are consumed; however there is limited information on the changes of polyphenolic content of the berries during juice processing. This study investigated the effects of three different pretreatments (grinding plus blanching; only grinding; only blanching) for cranberry juice processing on the concentrations of anthocyanins, flavonols, and procyanidins throughout processing. Flavonols and procyanidins were retained in the juice to a greater extent than anthocyanins, and pressing resulted in the most significant losses in polyphenolics due to removal of the seeds and skins. Flavonol aglycones were formed during processing as a result of heat treatment. Drying of cranberry pomace resulted in increased extraction of flavonols and procyanidin oligomers but lower extraction of polymeric procyanidins. The results indicate that cranberry polyphenolics are relatively stable during processing compared to other berries; however, more work is needed to determine their fate during storage of juices.